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1 Terminal Management System

Flask API for the Mellon Group DevOps Bootcamp final assignment. The local stack runs with Docker Compose and includes:

- tms-api: Flask application built from app/Dockerfile
- mysql: official MySQL image with persistent data and automatic schema/seed import
- redis: official Redis image used as the cache layer
- tms-cleanup: crontab-based cleanup worker for expired decommissioned terminals

The Flask entrypoint is app/main.py. The cleanup worker uses a crontab entry executed by super-cronic inside a separate container.

1.1 Project Layout

```
.
├── app/
│   ├── Dockerfile
│   ├── main.py
│   └── requirements.txt
├── cleanup/
│   ├── Dockerfile
│   ├── cleanup.sh
│   ├── crontab
│   └── entrypoint.sh
├── db/
│   └── init/
│       ├── 01_schema.sql
│       └── 02_seed.sql
├── docker-compose.yml
├── .env.example
└── README.md
```

1.2 Environment

Create a local .env file from the template:

```
cp .env.example .env
```

Update the password values before running the stack. The real `.env` file is ignored by git.

1.3 Run

```
docker compose up --build
```

The API will be available at:

```
http://localhost:5000
```

Useful endpoints:

- GET `/health`: checks both MySQL and Redis connectivity and returns 503 if either component is degraded
- GET `/schema/terminals`: runs the schema exploration query from the assignment
- GET `/terminals`: lists all terminals
- GET `/terminals?enabled=true`: lists enabled terminals
- GET `/terminals?enabled=false`: lists decommissioned/disabled terminals
- GET `/templates`: lists templates
- GET `/templates/<id>`: returns one template
- POST `/terminals/from-template`: creates a terminal from a template and merchant MID
- GET `/terminals/<tid>`: returns details for one terminal
- GET `/terminals/flagged`: lists terminals with a non-zero scenario number
- POST `/terminals/<tid>/flag`: sets `scenario_number` and updates `updated_on`
- POST `/terminals/<tid>/unflag`: sets `scenario_number` to 0 and updates `updated_on`
- POST `/terminals/<tid>/decommission`: disables a terminal and queues it for deletion after 3 days
- GET `/terminals/decommissioned`: lists terminals in the decommission queue
- GET `/statistics/by-hardware`: terminal count by hardware model
- GET `/statistics/by-state`: active/inactive terminal counts
- GET `/statistics/by-hardware-family`: terminal count by hardware family
- GET `/statistics/idle-distribution`: terminal count by idle-days bucket

Application logs are written to stdout with timestamp, level, and message. Database and Redis operations are wrapped with explicit error handling and return JSON error responses instead of silent failures.

1.4 Redis Caching

The API uses a cache-aside pattern with Redis. GET `/terminals` is cached for 30 seconds, and GET `/statistics/*` endpoints are cached for 60 seconds. Cache operations are best-effort: if Redis is unavailable, the API logs the Redis error and continues by reading from MySQL.

Every write endpoint clears the whole application cache before returning:

- POST `/terminals/<tid>/flag`
- POST `/terminals/<tid>/unflag`
- POST `/terminals/<tid>/decommission`
- POST `/terminals/from-template`

To verify cache behavior, watch the API logs:

```
docker compose logs -f tms-api
```

In another terminal:

```
curl http://localhost:5000/terminals
curl http://localhost:5000/terminals
```

```
curl -X POST http://localhost:5000/terminals/T0101001/flag \
-H "Content-Type: application/json" \
-d '{"scenario_number":"5"}'
curl http://localhost:5000/terminals
```

The expected log flow is Cache MISS, then Cache HIT, then cache clear after the write, then Cache MISS again.

1.5 Database Initialization

The SQL files in db/init/ are mounted to /docker-entrypoint-initdb.d inside the MySQL container. The official MySQL image executes them automatically only when the database volume is created for the first time.

To recreate the database from scratch:

```
docker compose down -v
docker compose up --build
```

1.6 Schema Note

The mid field belongs to the merchants table. The terminals table stores merchant_id, which references merchants.id; therefore a terminal's MID is retrieved through a join.

On API startup, the app applies the Feature A schema additions idempotently:

- Adds terminals.updated_on only if the column does not already exist.
- Creates decommission_queue only if the table does not already exist.

1.7 Feature A Examples

List terminals:

```
curl http://localhost:5000/terminals
curl "http://localhost:5000/terminals?enabled=true"
```

Terminal details:

```
curl http://localhost:5000/terminals/T0101001
```

Flag and unflag:

```
curl -X POST http://localhost:5000/terminals/T0101001/flag \
-H "Content-Type: application/json" \
-d '{"scenario_number":"5"}'
```

```
curl -X POST http://localhost:5000/terminals/T0101001/unflag
```

Decommission:

```
curl -X POST http://localhost:5000/terminals/T0101001/decommission
curl http://localhost:5000/terminals/decommissioned
```

1.8 Feature B Examples

List templates:

```
curl http://localhost:5000/templates
```

Template details:

```
curl http://localhost:5000/templates/1
```

Create terminal from template:

```
curl -X POST http://localhost:5000/terminals/from-template \  
-H "Content-Type: application/json" \  
-d '{"template_id":1,"mid":"MID000101"}'
```

The create operation runs in a single database transaction. It validates the template, validates the merchant MID, locks the merchant's existing terminals, calculates the next TID suffix, inserts the new terminal, and returns 201 Created.

1.9 Feature D Examples

Statistics endpoints use Pandas for aggregation and Redis cache with a 60-second TTL:

```
curl http://localhost:5000/statistics/by-hardware  
curl http://localhost:5000/statistics/by-state  
curl http://localhost:5000/statistics/by-hardware-family  
curl http://localhost:5000/statistics/idle-distribution
```

1.10 Tests

Run the unit tests with:

```
pytest -q
```

The tests use fake database and Redis objects, so they do not require Docker containers to be running.

1.11 CI/CD

GitHub Actions workflows live in `.github/workflows/`.

CI runs on pull requests and pushes to main or master. It:

1. Checks out the repository.
2. Sets up Python 3.12.
3. Installs dependencies from `app/requirements.txt`.
4. Runs unit tests with `pytest -q`.
5. Creates `.env` from `.env.example`.
6. Validates docker compose config.
7. Builds the `tms-api` Docker image.
8. Builds the `tms-cleanup` Docker image.

CD runs on pushes to main and can also be started manually from GitHub Actions. It:

1. Logs in to GitHub Container Registry with `GITHUB_TOKEN`.
2. Builds and pushes the API image:

```
ghcr.io/<owner>/tms-api:latest  
ghcr.io/<owner>/tms-api:<commit-sha>
```

3. Builds and pushes the cleanup image:

```
ghcr.io/<owner>/tms-cleanup:latest  
ghcr.io/<owner>/tms-cleanup:<commit-sha>
```

The CD workflow publishes images. A real server deployment step can be added later when the target host, SSH secrets, and deployment strategy are known.

1.12 Bonus Cron Cleanup

The tms-cleanup service runs in a separate container with a crontab file. Configure the schedule with:

```
CLEANUP_CRON=0 2 * * *
```

At container startup, cleanup/entrypoint.sh writes this crontab line to /etc/tms-cleanup.generated:

```
0 2 * * * . /etc/tms-cleanup.env; /usr/local/bin/cleanup.sh >> /proc/1/fd/1 2>> /proc/1/fd/2
```

The shell script deletes expired rows where decommission_queue.delete_after < NOW(). Because decommission_queue.tid has a foreign key to terminals.tid, it deletes in this order inside one transaction:

1. Delete from decommission_queue.
2. Delete the matching rows from terminals.

Run the cleanup once manually for verification:

```
docker compose run --rm -e CLEANUP_ONCE=true tms-cleanup
```

Inspect the installed crontab file:

```
docker exec tms-cleanup cat /etc/tms-cleanup.generated
```

Watch cron runner logs:

```
docker compose logs -f tms-cleanup
```

1.12.1 Cleanup File Details

cleanup/Dockerfile

Builds the image for the cleanup container. It uses the official mysql:8.4 image so the container has the correct MySQL 8 client and can authenticate against the MySQL server. It downloads supercrronic, copies the cleanup scripts into the image, and sets cleanup/entrypoint.sh as the container entrypoint.

cleanup/entrypoint.sh

Runs when the tms-cleanup container starts. It creates /etc/tms-cleanup.env from Docker environment variables, creates /etc/tms-cleanup.generated with the final crontab line, prints that crontab line to stdout, and starts supercrronic. If CLEANUP_ONCE=true is passed, it runs cleanup.sh once and exits instead of starting the cron runner.

cleanup/crontab

Contains the default crontab line:

```
0 2 * * * . /etc/tms-cleanup.env; /usr/local/bin/cleanup.sh >> /proc/1/fd/1 2>> /proc/1/fd/2
```

cleanup/cleanup.sh

Does the actual cleanup work. It validates required MySQL variables, selects expired TIDs from decommission_queue, deletes those rows from decommission_queue, then deletes the matching terminals from terminals. The delete order matters because decommission_queue.tid references terminals.tid with a foreign key. The script also clears Redis tms:* keys after a successful cleanup.

1.12.2 Crontab Details

The default schedule is:

```
0 2 * * *
```

Crontab fields are:

```
minute hour day-of-month month day-of-week
```

So `0 2 * * *` means:

- 0: at minute 0
- 2: at hour 2, meaning 02:00
- *: every day of the month
- *: every month
- *: every day of the week

Therefore the cleanup runs every day at 02:00 inside the cleanup container.

The full generated crontab command is:

```
0 2 * * * . /etc/tms-cleanup.env; /usr/local/bin/cleanup.sh >> /proc/1/fd/1 2>> /proc/1/fd/2
```

This does three things:

1. `. /etc/tms-cleanup.env` loads the MySQL and Redis environment variables for the cron process.
2. `/usr/local/bin/cleanup.sh` runs the cleanup script.
3. `>> /proc/1/fd/1 2>> /proc/1/fd/2` sends stdout and stderr to the container logs, so docker compose logs `tms-cleanup` shows cron output.

The generated environment file contains values like:

```
MYSQL_HOST='mysql'
MYSQL_PORT='3306'
MYSQL_DATABASE='tms'
MYSQL_USER='tms_app'
MYSQL_PASSWORD='...'
REDIS_HOST='redis'
REDIS_PORT='6379'
REDIS_DB='0'
```

Inspect the generated files inside the running container:

```
docker exec tms-cleanup cat /etc/tms-cleanup.generated
docker exec tms-cleanup cat /etc/tms-cleanup.env
```

1.13 Data Schema Check

Use this query to inspect the terminals table:

```
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_SCHEMA = DATABASE() AND TABLE_NAME = 'terminals'
ORDER BY ORDINAL_POSITION;
```

After the API starts, the result includes the assignment seed columns plus the `updated_on` column added by the application migration.